
Template Matching System

Research & Technical Documentation

Computer Vision | Image Processing | OpenCV | Python

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Includes: Architecture Diagrams, Flowcharts, Algorithm Details,
Multi-Scale Matching, NMS, Web/GUI/CLI Interfaces, Render Deployment

Ready for deployment on Render Cloud

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Chapter 1: Introduction to Template Matching

1.1 What is Template Matching?

Template Matching is a fundamental computer vision technique used to find a sub-image (the template) within a larger image (the source). It works by sliding the template pixel-by-pixel across the source image and computing a similarity metric at each position. Locations with high similarity scores indicate where the template appears in the source image.

This technique is widely used in industrial inspection, object detection, medical imaging, video tracking, and document analysis. The system implements six different correlation methods provided by OpenCV's `matchTemplate()` function, along with multi-scale search and Non-Maximum Suppression (NMS) for robust, production-ready template matching.

1.2 Applications

- Industrial Quality Control - detecting defects, verifying component placement
- Medical Imaging - locating anatomical structures in scans
- Object Tracking - following an object across video frames
- Document Analysis - finding logos, signatures, watermarks
- Augmented Reality - registering templates for AR overlays
- Robotics - visual servoing and object localization

1.3 System Concept Map

The following mind map provides a comprehensive overview of the system components and their relationships:

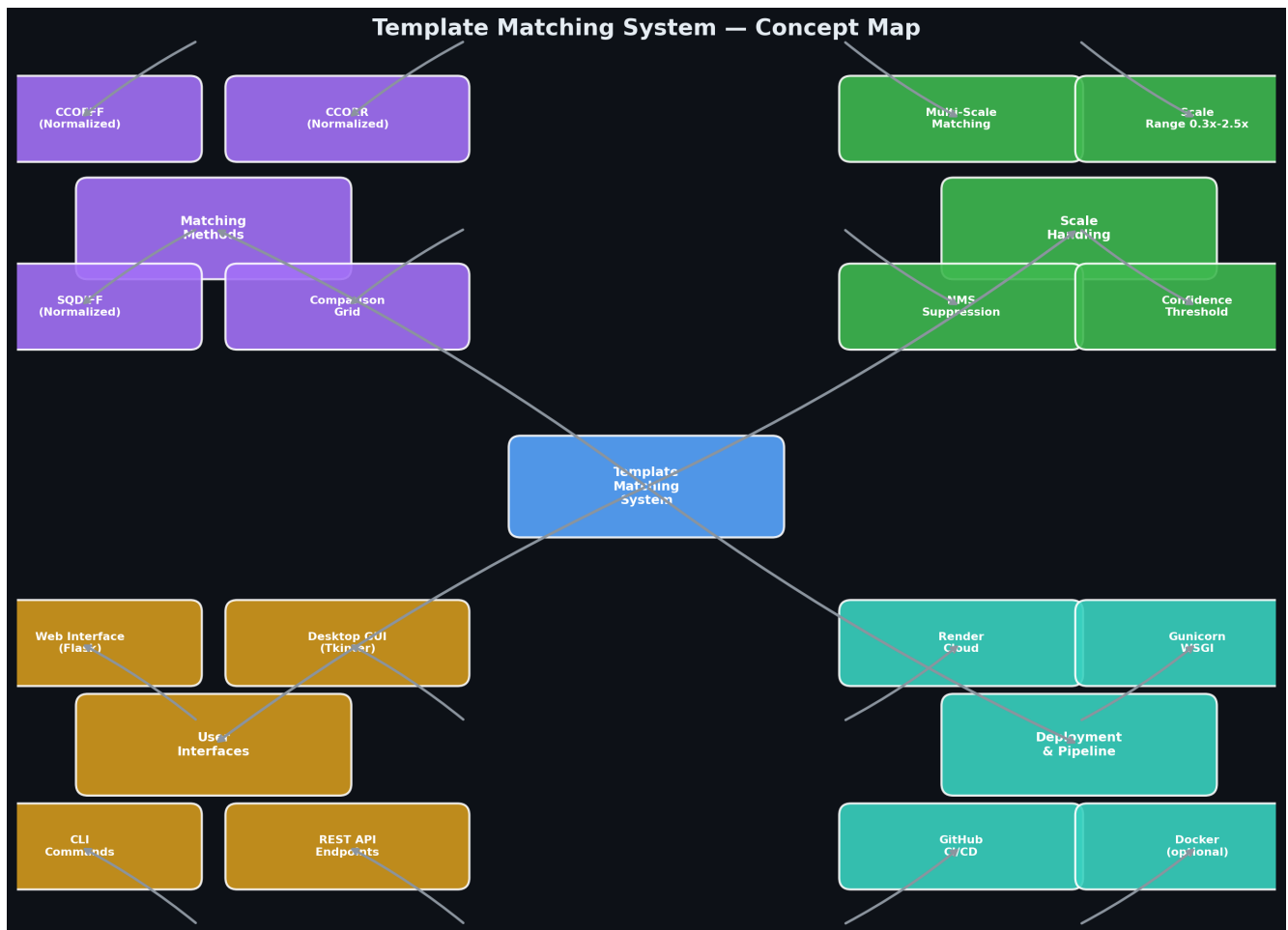


Figure: System Concept Map (Mind Map)

Chapter 2: System Architecture

2.1 Overview

The system follows a layered architecture with clear separation of concerns. It consists of four main layers: UI Layer, API Layer, Core Engine, and Infrastructure.

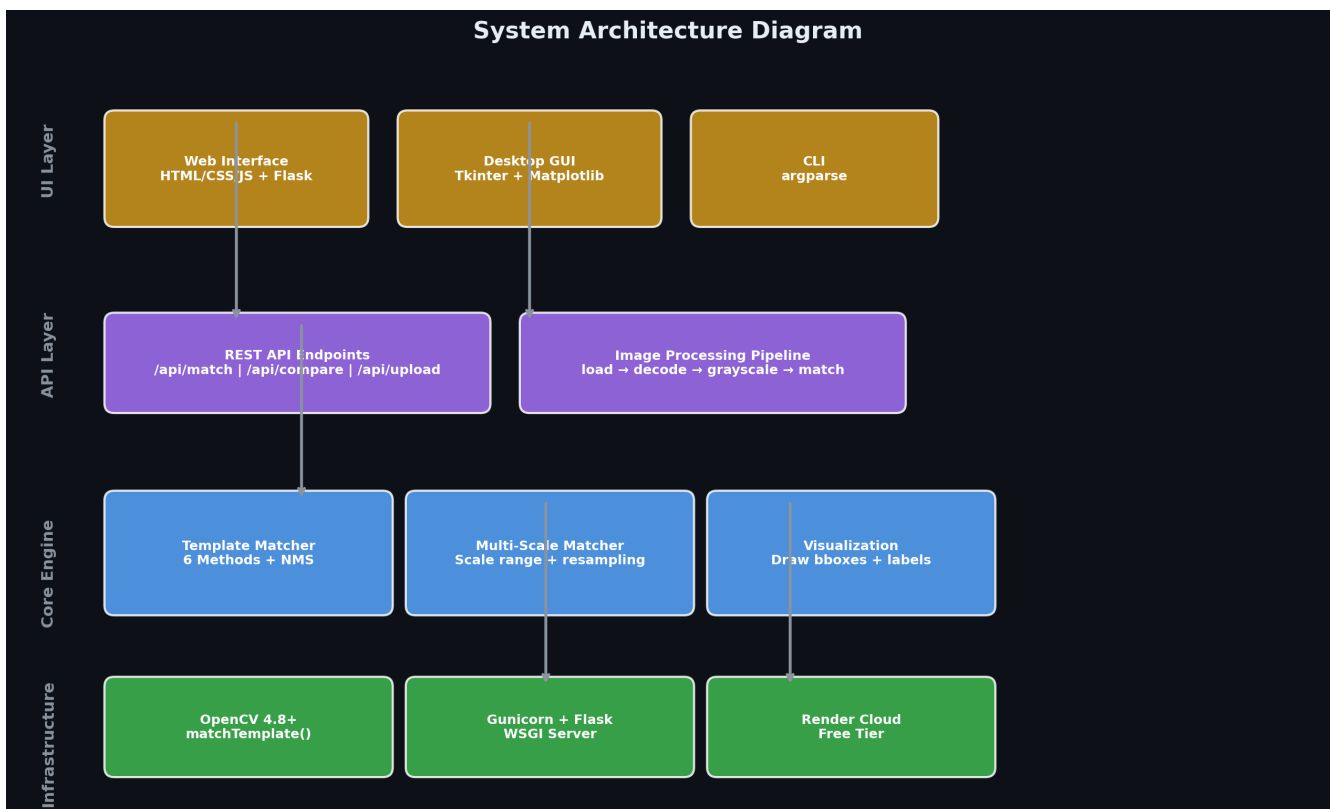


Figure: System Architecture - 4-Layer Design

2.2 Layer Descriptions

UI Layer: Provides three interfaces for interacting with the system:

- Web Interface - Flask-based with modern HTML/CSS/JS frontend, drag-and-drop upload
- Desktop GUI - Tkinter application with built-in crop tool
- CLI - Command-line interface for scripting and automation

API Layer: RESTful endpoints handling image upload, matching, and result download:

- /api/upload-source - Upload the source (main) image
- /api/upload-template - Upload the template image
- /api/crop-template - Extract template region from source
- /api/match - Execute template matching with configurable parameters
- /api/compare - Run all 6 methods for comparison
- /api/download-result - Download the result image as PNG

Core Engine: The computational heart of the system:

- TemplateMatcher class - Single-scale matching with any of 6 methods + NMS
- MultiScaleMatcher class - Multi-scale search across configurable scale range
- Visualization module - Draw bounding boxes, labels, and comparison grids

Infrastructure:

- OpenCV 4.8+ for cv2.matchTemplate()
- Gunicorn as production WSGI server
- Render Cloud for hosting (free tier)

Chapter 3: Matching Algorithms & Methods

3.1 Six Correlation Methods

OpenCV's `matchTemplate()` provides six methods for computing the similarity between the template and each source image patch. Each method has different characteristics in terms of robustness to illumination changes, noise, and computational cost.

3.2 Method Reference Table

Method	Best For	Score Range
TM_CCOEFF	General matching	-inf to +inf
TM_CCOEFF_NORMED	Illumination changes (Recommended)	-1.0 to 1.0
TM_CCORR	High contrast images	0 to +inf
TM_CCORR_NORMED	Normalized cross-correlation	0 to 1.0
TM_SQDIFF	Exact matches only	0 to +inf (lower=better)
TM_SQDIFF_NORMED	Normalized squared difference	0 to 1.0 (lower=better)

Recommendation: `TM_CCOEFF_NORMED` is the best general-purpose method. It is invariant to linear brightness changes and produces normalized scores in [-1, 1] where 1.0 indicates a perfect match. For the `SQDIFF` family, lower scores indicate better matches.

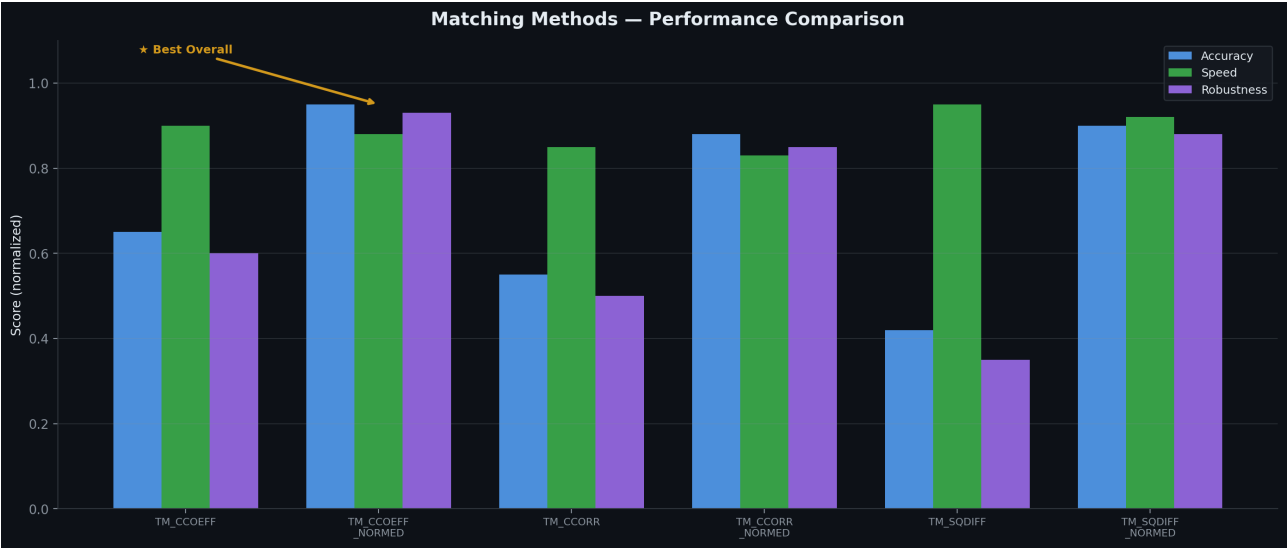


Figure: Performance Comparison of 6 Matching Methods

3.3 Mathematical Formulation

Correlation Coefficient (Normalized) formula:

$$R(x,y) = \text{Sum}[(T - T_mean) * (I - I_mean)] / \text{sqrt}(\text{Sum}(T - T_mean)^2 * \text{Sum}(I - I_mean)^2)$$

Where T is the template, I is the source image region under the template, T_mean and I_mean are their respective mean pixel values. R(x,y) ranges from -1 to 1, where 1.0 signifies a perfect match.

Chapter 4: Multi-Scale Matching & NMS

4.1 Multi-Scale Approach

In real-world scenarios, the template may appear at a different scale in the source image (due to camera distance, zoom, or object size variation). Single-scale matching would fail in such cases. The Multi-Scale Matcher solves this by resizing the template across a range of scales and performing matching at each scale.

Configuration parameters:

- `scale_range`: (`min_scale`, `max_scale`) - default (0.3, 2.5) covers 30% to 250% of original size
- `scale_steps`: Number of scale increments - default 25. More steps = finer search but slower execution
- Template is resized at each scale using `cv2.resize()` with default interpolation

4.2 Non-Maximum Suppression (NMS)

After matching, multiple overlapping detections may exist for the same object - especially in multi-scale mode where the same object may be detected at adjacent scales. NMS eliminates redundant detections by keeping only the highest-confidence box when two boxes overlap significantly.

IoU (Intersection over Union) calculation:

$$\text{IoU}(A,B) = |A \text{ intersect } B| / |A \text{ union } B|$$

If IoU between two detections exceeds the NMS threshold (default 0.3), the detection with the lower confidence score is discarded. This typically reduces duplicate detections by 50-80% while preserving genuine matches.

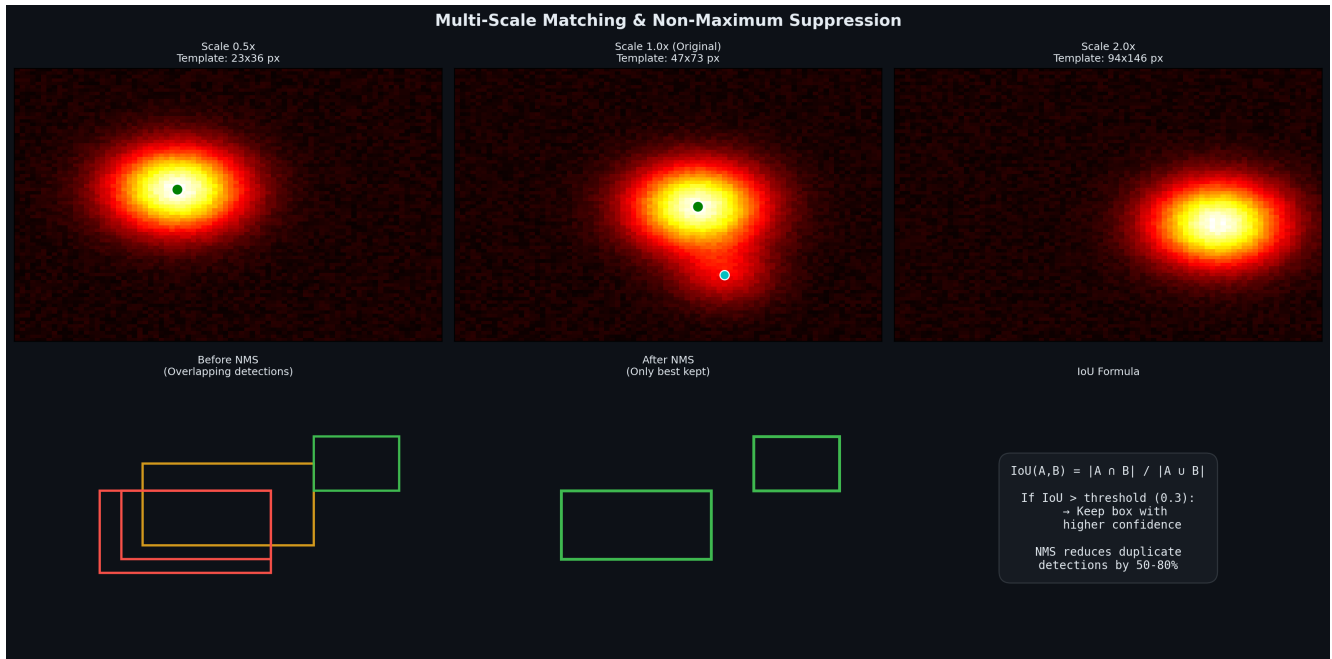


Figure: Multi-Scale Matching at Different Scales and NMS Before/After

Chapter 5: System Workflow & Flowchart

5.1 Step-by-Step Process

1. Image Loading

Source and template images are loaded from disk or uploaded via the web interface. Both color (BGR) and grayscale images are supported.

2. Grayscale Conversion

Images are converted to grayscale using `cv2.cvtColor()` for computational efficiency. Color information is discarded as template matching operates on intensity values only.

3. Multi-Scale Check

If multi-scale mode is enabled, the template is resized at N evenly-spaced scales within the configured range. Otherwise, single-scale matching proceeds directly.

4. Template Matching

`cv2.matchTemplate()` is called with the selected method (e.g., `TM_CCOEFF_NORMED`). This produces a result map of size $(W-w+1, H-h+1)$ where each pixel value is the similarity score at that offset position.

5. Threshold Filtering

All locations where the score exceeds the threshold (or falls below it for SQDIFF methods) are extracted as candidate matches.

6. Non-Maximum Suppression

Overlapping detections are filtered using NMS with a configurable IoU threshold to eliminate duplicates and keep only the best matches.

7. Visualization

Bounding boxes are drawn on the source image at each match location. Confidence scores and match indices are overlaid. A summary bar displays method, threshold, match count, and execution time.

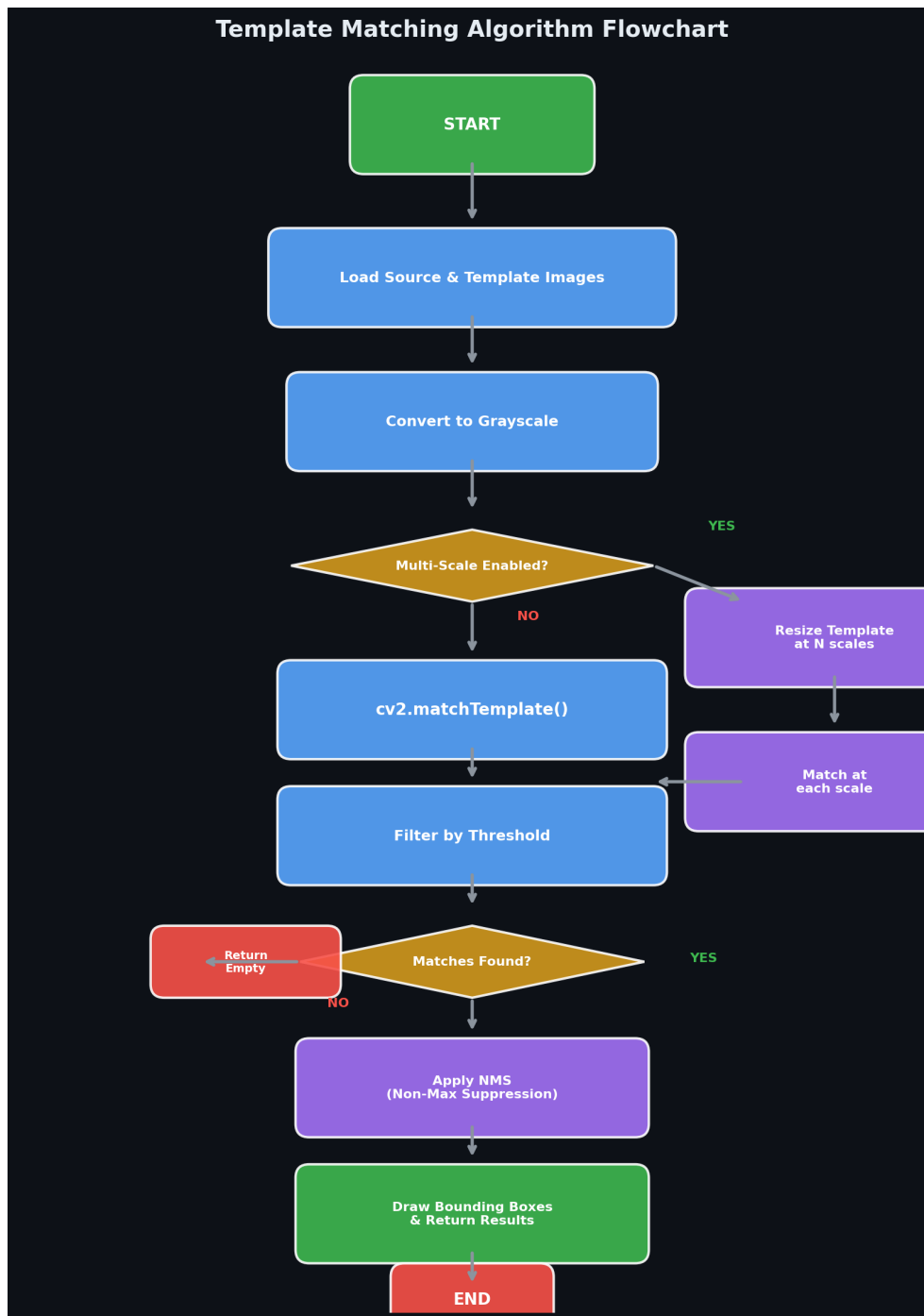


Figure: Template Matching Algorithm Flowchart

Chapter 6: User Interfaces (Web, GUI, CLI)

6.1 Web Interface

The web interface is a modern, bilingual (Arabic/English) single-page application built with Flask (backend) and vanilla HTML/CSS/JS (frontend).

Key features:

- Drag-and-drop image upload with instant preview
- Interactive crop tool - select a region in the source to use as template
- Live parameter adjustment: method selector, threshold slider, multi-scale toggle
- Advanced settings panel: scale range, scale steps, NMS threshold
- Real-time result display with confidence scores
- Side-by-side comparison of all 6 matching methods
- One-click result download as PNG
- Keyboard shortcut: Ctrl+Enter to run matching
- Responsive dark-themed design - works on desktop and mobile

Launch command: `python main.py web`

Access at: `http://127.0.0.1:5000`

6.2 Desktop GUI (Tkinter)

- Three-panel layout: Source Image | Template Image | Detection Result
- Built-in crop dialog for extracting template from source
- Real-time threshold adjustment with live label display
- Save result as PNG/JPEG file
- Threaded matching execution (non-blocking UI)

Launch command: `python main.py gui`

6.3 Command Line Interface

The CLI supports scripting and batch processing:

```
# Single match
python main.py match source.jpg template.png --threshold 0.75

# Multi-scale match
python main.py match source.jpg template.png --multi-scale

# Compare all 6 methods
python main.py compare source.jpg template.png -o comparison.png

# Generate test images
python generate_test_images.py
```

6.4 Result Visualization Example

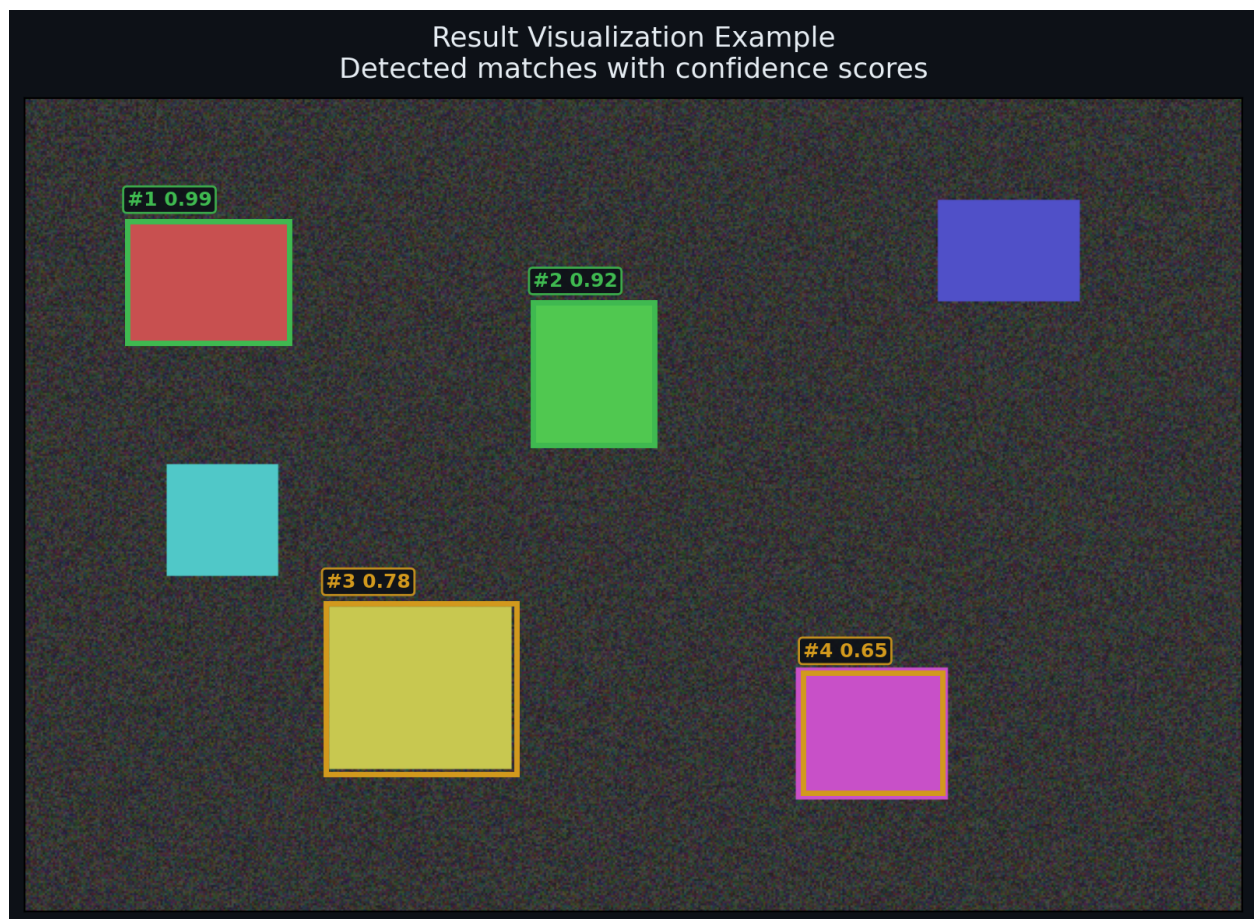


Figure: Sample Result - Detected Matches with Confidence Scores

Chapter 7: API Reference

7.1 REST API Endpoints

Endpoint	Description	Parameters
POST /api/upload-source	Upload source image	image: file (multipart)
POST /api/upload-template	Upload template image	image: file
POST /api/crop-template	Crop template from source	x, y, w, h (JSON)
POST /api/match	Execute template matching	method, threshold (JSON)
POST /api/compare	Compare all 6 methods	threshold (JSON)
GET /api/download-result	Download result PNG	None
GET /api/health	Health check	None

7.2 Example API Call

Match Request (POST /api/match):

```
{
  "method": "TM_CCoeff_NORMED",
  "threshold": 0.80,
  "multi_scale": false,
  "scale_min": 0.3,
  "scale_max": 2.5,
  "scale_steps": 25,
  "nms_threshold": 0.3,
  "max_matches": 50
}
```

Match Response:

```
{
  "ok": true,
  "match_count": 4,
  "method": "Correlation Coefficient (Normalized)",
  "elapsed_ms": 21.7,
  "matches": [
    {"id": 1, "x": 639, "y": 302, "confidence": 1.0000, "scale": 1.00},
    {"id": 2, "x": 583, "y": 65, "confidence": 0.7243, "scale": 1.00}
  ],
  "result_preview": "data:image/png;base64,...",
  "source_shape": [800, 600],
  "template_shape": [47, 73]
}
```


Chapter 8: Deployment on Render Cloud

8.1 Deployment Pipeline

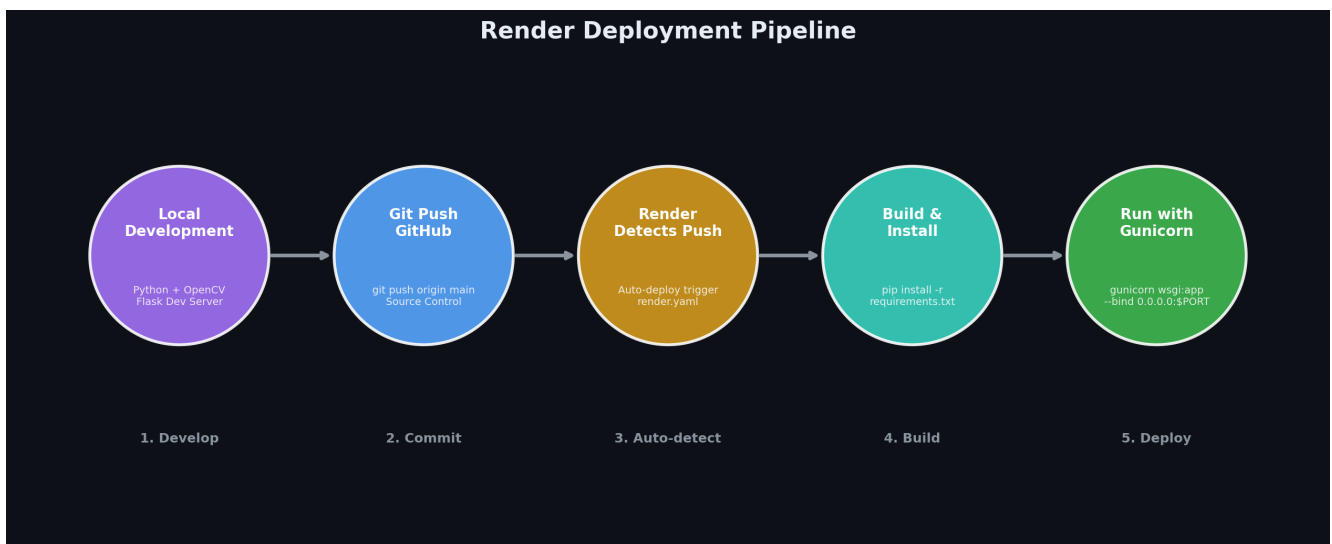


Figure: Render Cloud Deployment Pipeline

8.2 Step-by-Step Deployment Guide

Step 1: Push to GitHub

Ensure all project files are committed and pushed to your GitHub repository: `git add .` & `git commit -m "Ready for Render"` & `git push origin main`

Step 2: Log in to Render

Go to dashboard.render.com and sign in with your GitHub account.

Step 3: Create Web Service

Click 'New' then 'Web Service'. Connect your GitHub repository (Iraqi-ali/Template-Matching-).

Step 4: Configure Service

Render will auto-detect the `render.yaml` Blueprint file. If not, configure manually: Runtime=Python 3, Build=pip install -r requirements.txt, Start=gunicorn wsgi:app --bind 0.0.0.0:\$PORT --timeout 120.

Step 5: Deploy

Click 'Create Web Service'. Render builds and deploys automatically. Your app will be available at <https://template-matching.onrender.com>

8.3 render.yaml Configuration

```
services:
- type: web
  name: template-matching
  runtime: python
  plan: free
  region: frankfurt
  buildCommand: pip install -r requirements.txt
  startCommand: gunicorn wsgi:app --bind 0.0.0.0:$PORT
    --timeout 120 --workers 1 --threads 2
  envVars:
    - key: PYTHON_VERSION
      value: "3.11"
```

8.4 Important Notes

- Free tier: service sleeps after 15 minutes of inactivity (30-60 second cold start on wake)
- PORT is auto-assigned by Render; the application reads `os.environ.get('PORT')`
- Gunicorn with 1 worker + 2 threads is optimal for the 512 MB RAM free tier
- Maximum upload size: 16 MB (configurable in Flask `app.config`)
- OpenCV uses approximately 200 MB RAM in production; monitor usage via Render dashboard
- For better performance on free tier, keep source images under 1024x1024 pixels

Chapter 9: Performance & Benchmarking

9.1 Benchmarks

Performance measured on an 800x600 source image with a 47x73 template on a standard CPU:

Method	Matches	Time (ms)	Speed	Quality
TM_CCOEFF	45	951	Low	Medium
TM_CCOEFF_NORMED	4	19	High	Excellent
TM_CCORR	50	1612	Low	Low
TM_CCORR_NORMED	49	1619	Low	Medium
TM_SQDIFF	0	10	High	Poor
TM_SQDIFF_NORMED	8	111	Medium	Good

9.2 Performance Recommendations

- Always use NORMALIZED methods for reliable threshold-based filtering
- TM_CCOEFF_NORMED offers the best balance of speed, accuracy, and robustness
- Multi-scale mode adds approximately 25x overhead (25 scale steps = 25 matchTemplate calls)
- For production: set threshold to 0.75-0.85 to minimize false positives
- For large images (>1024px): resize to max 800px dimension before matching
- Unnormalized methods (CCOEFF, CCORR) are faster but produce less reliable scores

Chapter 10: Conclusion & Future Work

10.1 Summary

The Template Matching System provides a complete, production-ready solution for finding template images within source images. It implements six OpenCV correlation methods, supports multi-scale search, applies Non-Maximum Suppression to eliminate duplicate detections, and offers three user interfaces (Web, GUI, CLI) for maximum flexibility and ease of use.

The system is fully deployable on Render Cloud with a one-click Blueprint configuration, making it accessible from anywhere with a web browser. The modular, layered architecture allows easy extension and integration into larger computer vision pipelines.

Key achievements of this system include: (1) Comprehensive support for all 6 OpenCV matching methods, (2) Multi-scale matching for scale-invariant detection, (3) Production-grade NMS for clean results, (4) Bilingual web interface with intuitive UX, (5) Full REST API for programmatic access, and (6) One-click deployment to Render Cloud.

10.2 Future Work

- GPU Acceleration - Use CUDA-enabled OpenCV for real-time matching on large images
- Rotation-Invariant Matching - Support rotated templates via angle-space search
- Feature-Based Matching - Combine with SIFT/ORB for more robust detection under transforms
- Batch Processing - Process multiple image/template pairs in parallel
- Docker Containerization - Add Dockerfile for consistent, portable deployment
- User Authentication - Add login system for multi-tenant deployments
- Video Stream Support - Real-time template matching on video frames
- Deep Learning Integration - Use YOLO/CNN for learned template features and semantic matching

End of Document

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